

When Should Active RAG Retrieve? A Budget-Aware Evaluation of Utility, Calibration, and Cost

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Abstract

Active RAG systems decide when to retrieve external knowledge during generation, making them a budget-sensitive case of agentic RAG and self-adaptive retrieval. Yet evaluations often leave the operating point underspecified: two systems may both claim a 50% evidence-usage budget while realizing different held-out usage rates, so higher accuracy can reflect a looser budget rather than a better retrieval policy. We study budget-aware evaluation for Active RAG by recasting active retrieval as utility estimation, where retrieval is valuable only through its marginal correctness change over a no-retrieval answer. This view separates three questions that single-point evaluations conflate: whether trigger scores rank useful retrieval decisions, whether thresholds calibrated on past data meet future budgets, and how trigger-side computation changes deployment cost. We operationalize these questions with exact top- k utility frontiers, deployable threshold frontiers, conservative budget frontiers, harm audits, and cost decompositions. Across knowledge-intensive multi-hop QA datasets and open instruction models, retrieval harm is non-negligible, router rankings change across datasets and budgets, nominal thresholds can miss target usage, and simple uncertainty or retrieval-score baselines often rival learned utility routers. Budget-aware Active RAG evaluations should therefore report frontiers, realized usage, threshold-transfer error, harm rates, and cost decompositions alongside accuracy.

Keywords

Active RAG, retrieval-augmented generation, agentic RAG, budget-aware evaluation, retrieval budgets, utility estimation, calibration, cost accounting

1 Introduction

Knowledgeable foundation models increasingly need to decide when to rely on parametric memory and when to use retrieved evidence. Active RAG methods respond by asking when a model should retrieve or use evidence [2, 3, 13, 14, 24]. In practice, however, the phrase “when to retrieve” hides an operating-point problem. If two systems are both evaluated as adaptive RAG pipelines, but one uses retrieved evidence more often, an accuracy comparison alone does not tell us which system made better retrieval decisions.

The deployment question is therefore sharper: under a fixed evidence-usage or compute budget, which inputs deserve retrieval, and will the threshold chosen on a calibration split respect that budget on future inputs? This distinction is not cosmetic. In our 2,000-example Qwen2.5-1.5B experiments, routers calibrated to a nominal 50% evidence-usage target realize different held-out usage rates across random splits. On HotpotQA, common triggers exceed the target in 40%–100% of splits when any overshoot is counted. The overshoot magnitude is not always large, so we also report tolerance-based excess usage; the point is that a nominal target is not itself an observed budget.

We evaluate Active RAG as budgeted decision-making rather than as a single trigger-accuracy problem. This framing exposes four failure modes that unbudgeted comparisons can hide. First, *utility heterogeneity*: retrieval may help, be neutral, or harm. Second, *ranking failure*: a trigger score may not place beneficial cases above neutral or harmful cases. Third, *calibration failure*: a threshold that meets a budget on calibration data may miss it on held-out inputs. Fourth, *cost accounting failure*: evidence-usage rate is not total deployment cost, because uncertainty, probe-retrieval, and learned routers pay different generation, retrieval, and prompt-token costs.

Our contribution is to make these operating points comparable. We formulate active retrieval as budgeted utility estimation, where utility is the marginal correctness change from using retrieved evidence instead of a no-retrieval answer. We then operationalize an evaluation protocol that separates exact ranking quality, deployable threshold transfer, conservative budget behavior, retrieval harm, and heterogeneous trigger costs. Finally, we instantiate the protocol with representative trigger families—uncertainty scores, BM25 evidence scores, query-complexity triggers, and lightweight harm-aware utility routers—and audit them across datasets, model families, random splits, calibration sizes, manual and LLM correctness checks, and cost-aware simulations. The goal is not to crown a universal router, but to show how future Active RAG systems can be compared at explicit, reproducible operating points.

2 Related Work

Retrieval-augmented generation. RAG models combine generated text with evidence retrieved from an external corpus [15]. This improves knowledge-intensive tasks but introduces a cost-quality

trade-off: retrieval expands the prompt and can add irrelevant information. Recent efficient and on-device RAG studies make the same trade-off explicit at the system level, where retrieval quality, energy, and resource use must be considered together [4, 6].

Active and adaptive retrieval. FLARE retrieves during generation when low-confidence tokens indicate that upcoming content needs evidence [14]. Self-RAG trains models to emit reflection tokens that control retrieval and critique [2]. Adaptive-RAG selects among no-retrieval, single-step retrieval, and multi-step retrieval using query complexity labels [13]. DRAGIN estimates real-time information needs during generation [24], and UAR casts multiple active retrieval criteria as plug-and-play classification tasks [3]. Recent work also shows that uncertainty estimators remain competitive with more complex adaptive pipelines [21]. Adjacent agentic Text-to-SQL taxonomies frame LLM pipelines by inference-time autonomy and feedback loops [25, 33]. Related document-routed RAG work frames retrieval selection as a robustness-precision trade-off [5]. Our focus is orthogonal: regardless of the trigger family, thresholds should be calibrated and evaluated under explicit usage budgets.

Calibration. Calibration asks whether model scores correspond to empirical outcomes [10]. We use the term in a decision-theoretic sense: trigger scores are not final answers, but decision variables that should be thresholded on held-out data to meet a target usage rate.

3 Budgeted Utility Frontiers

The key distinction is between a *score* and an *operating point*. A trigger score can be useful if it ranks high-utility retrieval decisions above low-utility ones, but a deployed system also needs a threshold that respects a budget on future inputs. We therefore evaluate active retrieval through two linked questions: how good is the ranking, and how reliable is the calibrated threshold?

Let q_i be a question, y_i its gold answer, a_i^0 a no-retrieval answer, and a_i^R a retrieved-context answer. The net utility of using retrieval for that example is

$$u_i = \text{correct}(a_i^R, y_i) - \text{correct}(a_i^0, y_i), \quad (1)$$

where $u_i \in \{-1, 0, +1\}$ corresponds to harmful, neutral, or beneficial retrieval. More generally, an active RAG policy π chooses an action such as no retrieval, single-shot evidence use, or a more expensive evidence strategy. The deployment objective is

$$\max_{\pi} \mathbb{E}[\text{Acc}(a^{\pi(q)}, y)] \quad \text{s.t.} \quad \mathbb{E}[C(\pi, q)] \leq B. \quad (2)$$

In our binary experiments, $\pi(q) \in \{0, R\}$ and the usage budget counts examples whose final answer uses retrieved evidence.

Three frontiers. We distinguish three evaluation objects that are often conflated. The *exact frontier* sorts held-out examples by a router score and retrieves the top $\lfloor \rho n \rfloor$ examples at budget ρ ; it is not a deployable threshold, but diagnoses ranking quality under matched usage. The *deployable frontier* chooses a threshold on a calibration split and applies it unchanged to held-out inputs; it measures threshold transfer and reports realized usage. The *conservative frontier* uses a stricter threshold chosen with finite-sample slack; it

measures how much accuracy is lost when budget violations are discouraged.

Cost model. Our budget counts examples that use retrieved evidence in the final answer, not a universal wall-clock or monetary cost. Because trigger families pay different pre-decision costs, we account for each policy type separately:

$$\begin{aligned} C_m(q) = & C_m^{\text{pre}}(q) \\ & + (1 - \pi_m(q))C_m^{\text{skip}}(q) \\ & + \pi_m(q)C_m^{\text{use}}(q), \end{aligned} \quad (3)$$

where m indexes the router family. Query-only routers have a cheap query feature cost before the decision; skipped examples then pay for a no-retrieval answer, while triggered examples pay for retrieval, retrieved-context prompt tokens, and retrieved-context generation. Uncertainty routers use the no-retrieval generation itself as the pre-decision computation, so skipped examples reuse that answer and triggered examples pay for retrieval plus a second generation. Retrieval-score routers pay a probe retrieval before the decision; C_{probe} denotes this retrieval and can be reused by triggered examples. Full-feature BUR variants combine the no-retrieval generation and probe retrieval before deciding. This decomposition is narrower than a full deployment latency model, but it makes the operating point explicit and avoids conflating trigger computation with retrieved-context answering. This cost-aware view is also related to sequential filtering analyses that optimize decision pipelines under cost and selectivity assumptions [22].

Reference utility routers. We evaluate several router families under this protocol rather than claiming a single dominant trigger. The lightweight Budgeted Utility Router (BUR) is a reference learned router: it assigns a scalar score $s(q)$ that estimates expected net retrieval utility. The linear version trains a multinomial logistic regression over $u_i \in \{-1, 0, +1\}$ and scores examples as $P(u_i = +1) - P(u_i = -1)$. We also include a small gradient-boosted BUR variant and feature ablations to test when utility modeling helps relative to simple uncertainty, retrieval-score, and query-complexity baselines.

Budget-safe calibration. Nominal thresholds may violate the target budget on deployment inputs. We therefore evaluate a conservative quantile rule: instead of calibrating to ρ , choose the threshold for $\rho_{\text{safe}} = \max(0, \rho - \epsilon)$, where

$$\epsilon = \sqrt{\frac{\log(2/\delta)}{2n_{\text{cal}}}}. \quad (4)$$

Under the usual exchangeability assumption, this DKW-style slack motivates population usage control, but we treat it empirically: conservative thresholds should reduce held-out budget violations, possibly at the cost of accuracy. We report both nominal and conservative thresholds.

4 Experimental Setup

Datasets. We evaluate on three multi-hop QA benchmarks with candidate evidence paragraphs: HotpotQA distractor [31], 2WikiMultiHopQA [12], and MuSiQue [28]. Our main Qwen2.5-1.5B evaluation uses 2,000 validation examples per dataset with fixed random

Dataset	No	Full	Ben.	Harm	Neut.	Net u
2Wiki	21.3	28.5	15.9	8.8	75.3	7.1
HotpotQA	15.4	43.4	31.9	4.0	64.0	27.9
MuSiQue	2.2	11.9	11.2	1.5	87.3	9.7

Table 1: Base rates for Qwen2.5-1.5B on 2,000 examples per dataset. Values are percentages. No is no-retrieval accuracy; Full is always-retrieve accuracy; Ben. is the fraction where retrieval fixes a no-retrieval error; harm is the fraction where retrieval changes a correct answer to an incorrect one.

seeds, filtering MuSiQue to answerable examples. This controlled setting isolates the trigger decision from large-scale indexing: retrieval is BM25 over the candidate paragraphs for each question. We interpret these main results as controlled routing diagnostics rather than full open-domain RAG performance. To test whether the conclusions survive noisier evidence, we also run HotpotQA global-candidate BM25 and dense-retrieval diagnostics: one index is built over all 66,635 deduplicated validation candidate paragraphs, and each query retrieves from that shared pool rather than from its own gold candidate set. This adds cross-example retrieval noise, but is still not a full Wikipedia index.

Models and retrieval. Our main cross-dataset experiments use Qwen2.5-1.5B-Instruct [23] on 2,000 examples per dataset. We also run SmolLM2-1.7B-Instruct on HotpotQA and Granite-3.1-2B-Instruct on HotpotQA plus 2Wiki as non-Qwen family checks [1, 9]. For each example, the model first answers from memory. We then retrieve the top five BM25 paragraphs and ask the same model to answer using only those passages. Answers are scored with exact match or token F1 above 0.80 after standard normalization.

Triggers. We compare seven trigger scores spanning four families. An *Adaptive-RAG-style* baseline uses only the question text and cheap query-complexity features, approximating query-level complexity routing rather than reproducing the original system. We also evaluate *mean entropy*, the average entropy over generated no-retrieval tokens; *minimum token confidence*, one minus the minimum maximum-token probability; *BM25 top score*; *BM25 margin*, the difference between the top two BM25 scores; *BUR-linear*; and *BUR-GBM*. BUR features include generation uncertainty, BM25 scores, answer length, and question length. We split each dataset evenly into calibration and held-out test sets, stratified by utility class when possible. Thresholds are selected only on the calibration split. The main tables use a CPU-only five-split evaluation over budgets {10, 25, 50, 75, 90}%, which reports mean accuracy, usage error, violation rate, exact frontiers, and conservative frontiers.

5 Results

We now ask the sequence of questions a deployer would have to answer before trusting an Active RAG trigger. Is retrieval utility actually heterogeneous? Do scores rank useful retrieval decisions? Does a calibrated threshold meet the future budget? Do the conclusions survive model and evidence changes? And does evidence usage still tell the right story once trigger costs are counted?

RQ1: Is retrieval utility heterogeneous? If retrieval only helped, Active RAG would mostly be a cost-saving problem: retrieve whenever the budget allows. Table 1 shows that the decision is more

delicate because retrieval is not monotonic. For Qwen2.5-1.5B, always retrieving improves HotpotQA accuracy from 15.4% to 43.4%, while retrieval benefit occurs on 31.9% of examples. The same model sees a much smaller gain on 2Wiki, from 21.3% to 28.5%, because retrieval harm is also higher at 8.8%. MuSiQue is the hardest setting in absolute accuracy: no-retrieval answers are almost always wrong, but candidate retrieval raises accuracy from 2.2% to 11.9%.

This heterogeneity also changes with the evidence source. Table 2 repeats the diagnostic under noisier retrieval. Switching from candidate BM25 to the global HotpotQA BM25 index lowers always-RAG accuracy from 43.6% to 34.6%, increases harm from 3.4% to 5.8%, and lowers the best exact-50 router from 34.8% to 30.0%. A dense E5-small retriever recovers part of that gap, raising always-RAG to 38.6% and lowering harm to 3.8%, but still changes the best exact-50 router and remains below the per-example candidate setting. This is still not full Wikipedia retrieval, but it demonstrates that evidence noise and retriever family change both utility and router ranking.

RQ2: Do routers rank useful retrieval decisions consistently? Once retrieval has mixed utility, an active trigger is useful only if its score orders examples well. Table 3 shows that this ranking problem is not solved by any single trigger family. On HotpotQA, BUR-linear reaches 33.8% deployable accuracy at the 50% target, minimum token confidence reaches 33.3%, entropy reaches 32.9%, and the query-only Adaptive-RAG-style baseline reaches 31.8%. These numbers are close enough that a single operating point would invite over-interpretation. Their realized usage rates also differ, and their split-level violation rates range from 40% to 100% when any overshoot counts, although tolerance-based overshoot rates distinguish small finite-split excesses from larger misses. Under an exact held-out 50% top- k diagnostic, BUR-linear and minimum confidence both reach 32.7%, entropy reaches 32.5%, and the oracle reaches 45.9%. This gap is itself a key reason to report realized usage and exact diagnostics rather than only target budgets. On 2Wiki, query-complexity and uncertainty scores are strongest around the same target, while BM25 margin remains competitive and the confidence intervals overlap. On MuSiQue, the query-only Adaptive-RAG-style baseline is strongest at the 50% target, while entropy and BUR variants remain close under exact usage. These results support recent findings that uncertainty-based methods are strong baselines [21] and reinforce our main claim: the protocol should not depend on one router family winning.

Retrieval	Full	Ben.	Harm	Rand	Best@50	Oracle
Cand. BM25	43.6	33.4	3.4	28.8	Adapt. 34.8	47.2
Global BM25	34.6	26.8	5.8	24.4	BUR-lin 30.0	40.8
Global dense	38.6	28.8	3.8	26.4	Dense 30.8	42.8

Table 2: HotpotQA retrieval-source diagnostic ($n = 500$; no-RAG accuracy is 13.6% for all rows). Full is always-retrieve accuracy; Ben. and Harm are utility rates; Rand and Oracle are exact-50 accuracies; Best@50 gives the best router and exact-50 accuracy. The harder global settings change both utility and router ranking.

Setting	Router	Deploy acc.	Usage	Err.	Split viol.	Exact acc.
2Wiki / Qwen2.5-1.5B	Adapt.	26.2±0.4	49.9	1.9	20.0	26.4±0.4
	Entropy	24.9±0.7	49.9	1.2	60.0	25.0±0.9
	MinConf	25.7±0.9	49.4	1.1	20.0	25.8±0.9
	BM25-top	24.9±1.1	48.9	2.0	20.0	24.9±1.2
	BM25-mar	25.3±0.8	49.8	1.3	40.0	25.4±0.7
	BUR-lin	25.1±0.9	48.5	2.1	20.0	25.0±0.9
	BUR-GBM	25.0±0.7	48.7	2.6	20.0	25.0±0.8
HotpotQA / Qwen2.5-1.5B	Adapt.	31.8±1.4	49.3	2.3	40.0	31.9±1.3
	Entropy	32.9±0.9	51.3	1.3	100.0	32.5±0.7
	MinConf	33.3±1.6	51.5	2.9	60.0	32.7±0.9
	BM25-top	31.5±1.2	49.1	2.3	40.0	31.8±0.8
	BM25-mar	30.1±1.2	50.2	1.7	80.0	30.0±1.1
	BUR-lin	33.8±1.1	51.6	2.1	60.0	32.7±0.9
	BUR-GBM	33.3±1.4	51.3	2.2	80.0	32.8±0.9
MuSiQue / Qwen2.5-1.5B	Adapt.	9.9±0.8	49.7	1.1	40.0	9.8±0.6
	Entropy	8.5±0.6	51.3	2.8	60.0	8.4±0.2
	MinConf	8.3±0.7	51.7	2.6	80.0	8.1±0.4
	BM25-top	7.7±0.1	49.8	2.1	60.0	7.7±0.2
	BM25-mar	8.0±0.3	50.7	2.1	60.0	7.9±0.2
	BUR-lin	8.6±0.4	51.9	1.9	100.0	8.3±0.4
	BUR-GBM	8.5±0.4	51.0	1.4	60.0	8.4±0.3

Table 3: Five-split 50% budget diagnostics for Qwen2.5-1.5B on 2,000 examples per dataset. Deploy acc. uses a calibration-selected threshold applied to held-out examples. Usage is realized held-out evidence usage, Err. is absolute usage error, Split viol. is the fraction of random splits exceeding the target by any amount, and Exact acc. enforces strict held-out top- k usage.

RQ3: Do calibrated thresholds meet future budgets? The previous result separates ranking quality from a nominal target. The next question is whether a threshold chosen on calibration data behaves like the same budget on held-out inputs. Table 3 shows that this transfer is imperfect even in the controlled setting. In the five-split 2,000-example evaluation, nominal 50% thresholds violate the target in 28.6% of selected router-split combinations for 2Wiki/Qwen2.5-1.5B, 65.7% for HotpotQA/Qwen2.5-1.5B, and 65.7% for MuSiQue/Qwen2.5-1.5B. The failure is not that all routers retrieve too much; rather, target budgets, realized usage, and accuracy must be reported together.

Budget-safe calibration illustrates the corresponding trade-off. On HotpotQA, BUR-linear’s mean 50% target usage drops from 51.6% under nominal thresholding to 46.5% under the conservative rule, with accuracy moving from 33.8% to 32.0%. The rule is conservative rather than free: for the query-only Adaptive-RAG-style baseline on HotpotQA it reduces usage from 49.3% to 44.9% and accuracy from 31.8% to 30.6%. This illustrates the practical trade-off between budget violation and answer quality.

To check whether automatic harm labels reflect real failures, we manually annotate 60 retrieval-harm candidates. Fifty-two are valid harms. Most valid harms are comparison errors or distractor-entity switches, and annotators usually attribute them to generation over retrieved evidence rather than to obviously wrong retrieval. Thus retrieval harm is not only an indexing problem: even relevant evidence can pull the generator toward the wrong relation or entity. Because all utility labels depend on automatic answer correctness, we also run an expanded LLM-judge audit over 733 sampled rows across datasets, model families, and retrieval settings, following

the broader use of LLMs as evaluators while treating their judgments as audit evidence rather than ground truth [16]. The judge validates 93.3% of automatic benefit labels and 83.8% of automatic harm labels, while near-threshold EM/F1 cases are much noisier at 40.0% validity. We therefore treat benefit/harm rates as meaningful but report metric-threshold robustness rather than assuming all automatic labels are exact, following broader concerns that benchmark verdicts can shift with configuration choices, detectable-effect budgets, and accuracy-only summaries [11, 18, 26, 29, 30, 34].

RQ4: Do rankings persist across model families? If thresholds are operating decisions, they should not be assumed to behave identically across model families. We therefore compare the same evidence-usage frontier protocol across Qwen, SmolLM2, and Granite settings.

Table 4 checks whether the phenomenon persists across model families rather than only in the main Qwen setting. The clean 2,000-example SmolLM2 run has a lower no-retrieval baseline than the main Qwen run (9.8% vs. 15.4%), but retrieval still raises accuracy to 30.6%, with 23.5% beneficial cases and 2.6% harmful cases. At exact 50% usage, its best router reaches 23.4% compared with a 20.0% random policy; the oracle is 33.1%. Granite-2B gives the same qualitative picture at clean scale: retrieval raises accuracy from 13.6% to 33.2%, but 6.2% of examples are harmful; its best exact-50 router reaches 28.7% compared with 23.7% random and a 39.7% oracle. Qwen2.5-1.5B reaches 32.7% at exact 50% usage, compared with 27.9% random and a 45.9% oracle. Across the clean 2,000-example rows, retrieval utility is non-monotonic, and the best exact-50 router changes across models. Thus the calibration protocol transfers more cleanly than any individual router.

Model	n	No	Full	Benefit	Harm	Best@50	Acc.	Rand	Oracle
Qwen2.5-1.5B	2000	15.4	43.4	31.9	4.0	BUR-lin	32.7	27.9	45.9
SmolLM2	2000	9.8	30.6	23.5	2.6	BUR-GBM	23.4	20.0	33.1
Granite-2B	2000	13.6	33.2	25.8	6.2	BUR-GBM	28.7	23.7	39.7

Table 4: HotpotQA model-family diagnostic with explicit sample sizes. Full is always-retrieve accuracy; Best, Rand, and Oracle are exact held-out top- k accuracies at 50% evidence usage. Qwen2.5-1.5B, SmolLM2, and Granite use clean 2,000-example runs.



Figure 1: Accuracy versus estimated token-equivalent cost normalized by always-RAG. Evidence-usage budgets do not fully determine deployment cost because trigger families pay different pre-decision costs.

We also run a CPU-only multi-split calibration-size diagnostic after generation is complete. Across five random splits and calibration sizes from 32 to 250, nominal 50% thresholds violate the held-out budget in roughly half of method–setting combinations, whereas budget-safe thresholds nearly eliminate violations but retrieve conservatively. We also report tolerance-based overshoot rates, because a one-example excess and a 5-point excess should not be interpreted the same way. This supports treating budget

behavior as a first-class deployment metric rather than a table footnote.

RQ5: How does cost accounting change the conclusion? Evidence usage is a convenient budget, but it is not the full cost of an active policy. Figure 1 simulates heterogeneous trigger costs with tokenizer-aware prompt accounting. The accounting separates query-only routing, no-retrieval generation, probe retrieval, and retrieved-context generation. It shows why evidence usage alone is incomplete: on HotpotQA, the clean 2,000-example cost simulation shows that BUR-linear can look attractive under an evidence-usage budget while paying both no-RAG generation and probe retrieval before many retrieved-context generations. In contrast, query-only routing avoids those trigger-side costs, and a three-stage cascade reduces probe retrieval relative to BUR-linear. The cascade improves the 2Wiki accuracy-cost trade-off but is not uniformly best. This supports reporting cost decompositions rather than only retrieval rates.

6 Discussion

The central lesson is methodological. Active RAG triggers are not merely architectural switches; they are policies over uncertain future utility under a budget. Evaluating them only at one adaptive accuracy number can obscure whether a method is genuinely better, better ranked but poorly calibrated, or simply using retrieved evidence more often. Exact frontiers and deployable frontiers answer different questions: the former measure ranking quality at a fixed usage rate, while the latter measure whether a calibrated threshold transfers to held-out inputs. Cost-aware accounting then asks whether the chosen evidence-usage budget is the right proxy for deployment cost. Reporting all three makes the operating point explicit.

Our experiments are deliberately lightweight. The main Qwen2.5-1.5B runs use 2,000 examples per dataset, and the SmolLM2 and Granite HotpotQA family checks also use 2,000 examples; the global-candidate retrieval diagnostics remain smaller. The cost simulation also uses the clean 2,000-example Qwen2.5-1.5B setting, but remains a token-equivalent accounting diagnostic rather than a deployment latency measurement. We still rely mostly on candidate paragraph retrieval rather than a full Wikipedia index, and on small open instruction models. These choices make the study reproducible on a single consumer GPU, but they do not establish broad generality. Future work should add full-corpus retrieval, semi-structured retrieval with adaptive fusion and reranking [27], 7B/API-scale instruction models, multimodal recommendation-style retrieval with MLLM graph refinement [7], and long-form or long-context reasoning tasks where retrieval timing and context compression jointly shape the budget [8]. It should also evaluate model-side distillation,

which has reduced inference requirements in forecasting settings [17, 19], as a complementary efficiency axis alongside retrieval-side budget allocation.

7 Conclusion

We presented Active RAG as budgeted utility estimation rather than an unbudgeted pipeline choice. This perspective turns a vague question—whether to retrieve—into three measurable questions: which examples should receive retrieval, whether the calibrated threshold respects the future budget, and how much computation the trigger itself consumes. Across HotpotQA, 2Wiki, and MuSiQue, retrieval benefit, retrieval harm, router rankings, budget behavior, and cost rankings all vary substantially. Future Active RAG work should therefore report exact and deployable frontiers, realized usage, threshold-transfer error, harm rates, and cost decompositions alongside end-to-end task accuracy.

Limitations

This study covers three multi-hop QA datasets and small open instruction models, but still uses dataset-provided candidate paragraphs for the main cross-dataset runs. The HotpotQA global-candidate BM25 and dense diagnostics add cross-example retrieval noise, but they are not true full-Wikipedia open-domain indexes; absolute accuracies should therefore not be compared to full QA systems. Non-Qwen family checks remain limited: SmolLM2 is evaluated on HotpotQA, and Granite is evaluated on HotpotQA plus 2Wiki. Correctness is automatic and may under-credit paraphrases, although our manual harm audit, expanded LLM-judge correctness audit, and metric-threshold robustness diagnostic check the most important error classes. We do not yet evaluate long-form generation, multi-step retrieval, memory-augmented agents [20], API-scale models, or official Self-RAG/FLARE reproductions inside the same frontier protocol; the comparisons should therefore be read as representative trigger-family audits rather than faithful rankings of named Active RAG systems. Finally, our cost model separates trigger computation from retrieved-context answering, but omits deployment-side latency, dollar, and energy costs [32].

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